

CLASS :
EXAM NO :

TIME : 3 Hrs
MARKS : 80

Class 11 Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

SECTION A: MCQs (1 mark each) (14 × 1 = 14)

1) Which of the following statements correctly describes the characteristics of Ctenophores?

- a) They are radially symmetrical, triploblastic organisms with cellular level of organization
- b) They are radially symmetrical, diploblastic organisms with tissue level of organization
- c) They are bilaterally symmetrical, diploblastic organisms with tissue level of organization
- d) They are asymmetrical, diploblastic organisms with organ level of organization

2) Assertion: Ctenophores are commonly known as comb jellies.

Reason: They possess eight external rows of ciliated comb plates that aid in locomotion.

- a) Both assertion and reason are true, and reason is the correct explanation of the assertion
- b) Both assertion and reason are true, but reason is not the correct explanation of the assertion
- c) Assertion is true, but reason is false
- d) Both assertion and reason are false

3) Which of the following features distinguishes Ctenophores from other phyla?

- a) Presence of nematocysts for capturing prey
- b) Presence of bioluminescence
- c) Intracellular digestion only
- d) Separate sexes with indirect development

4) Which of the following is NOT a characteristic feature of Ctenophora?

- a) Exclusively marine habitat
- b) Both extracellular and intracellular digestion
- c) Presence of a water vascular system
- d) Tissue level of organization

5) In a leaf, the axillary bud is located:

- a) At the tip of the lamina
- b) At the junction of leaf and stem
- c) Between the veins of the leaf

d) At the base of the petiole

6) Assertion: The seed of castor is called an endospermic seed.

Reason: In castor, the endosperm formed as a result of double fertilization persists in the mature seed.

a) Both assertion and reason are true, and reason is the correct explanation of the assertion

b) Both assertion and reason are true, but reason is not the correct explanation of the assertion

c) Assertion is true, but reason is false

d) Both assertion and reason are false

7) When examining the external morphology of a plant part, you observe nodes, internodes, and axillary buds. This plant part is most likely a:

a) Root

b) Stem

c) Leaf

d) Flower

8) A flower with inferior ovary would have which of the following arrangements?

a) Hypogynous

b) Perigynous

c) Epigynous

d) Amphigynous

9) Based on the given figure of a floral diagram, identify the correct statement:

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a) The flower belongs to family Solanaceae

b) The flower has valvate aestivation of corolla

c) The flower has a superior ovary

d) The flower has zygomorphic symmetry

10) In a monocotyledonous seed, the embryo typically has:

a) Two cotyledons and endosperm

b) One cotyledon and no endosperm

c) One cotyledon and endosperm

d) Two cotyledons and no endosperm

11) Case Study: A botanist examines a flower with the following characteristics: radial symmetry, 5 united sepals, 5 free petals with twisted aestivation, numerous stamens fused to form a staminal tube, and a superior ovary with axile placentation.

Which family does this flower most likely belong to?

a) Solanaceae

b) Fabaceae

c) Malvaceae

d) Liliaceae

12) A plant has compound leaves where leaflets are arranged on either side of an axis. The leaf is:

a) Palmately compound

b) Pinnately compound

c) Simple with deep lobes

d) Whorled compound

13) In which type of placentation are ovules attached to the central axis formed by the fusion of septa?

a) Axile

b) Free central

c) Parietal

d) Marginal

14) Which of the following regions of a root is responsible for absorption of water and minerals from the soil?

a) Region of meristematic activity

b) Region of elongation

c) Region of maturation with root hairs

d) Root cap

SECTION B: Very Short Answer (2 marks each) ($10 \times 2 = 20$)

15) Compare the anatomical features that distinguish a monocot stem from a dicot stem with reference to the arrangement of vascular bundles.

16) How does the structure of dorsiventral leaf contribute to its photosynthetic efficiency?

Analyze the arrangement of palisade and spongy parenchyma.

17) Explain the functional significance of casparian strips in the endodermal cells of roots.

18) Differentiate between open and closed vascular bundles, and explain why this distinction is important in plant taxonomy.

19) Compare the structure and function of bulliform cells in monocot leaves with the typical epidermal cells in dicot leaves.

20) Analyze how the internal structure of a leaf is adapted differently in monocots versus dicots to support their respective venation patterns.

21) Explain the relationship between the arrangement of tissues in the stele of a root and its primary functions of absorption and conduction.

22) How does the anatomy of seed structure in monocots differ from dicots in terms of food storage and embryo position? What evolutionary advantage might this provide?

23) Compare the structural organization of ground tissue system in stems versus roots, explaining how these differences relate to their different functions.

24) Explain how the structure of stomatal apparatus in grasses differs from that in dicotyledonous plants, and discuss the adaptive significance of these differences.

SECTION C: Short Answer Type I (3 marks each) ($7 \times 3 = 21$)

25) Explain the adaptive significance of bulliform cells in grass leaves, and describe how their structural modifications enable plants to respond to water stress conditions. Relate this to the overall anatomy of monocot leaves.

26) Compare the structure and function of the endodermis in dicot roots with the starch sheath in dicot stems. How do these specialized layers contribute to the physiological processes of the plant?

- 27) Analyze how the arrangement of tissues in dorsiventral leaves maximizes photosynthetic efficiency. Explain the functional relationship between the palisade and spongy mesophyll in relation to gas exchange and light absorption.
- 28) Describe how the structural differences between monocot and dicot vascular bundles relate to their capacity for secondary growth. Why is this difference significant for plant development and survival?
- 29) Evaluate the ecological importance of stomatal apparatus in plants. How does the specialized structure of guard cells enable plants to balance the conflicting needs of gas exchange and water conservation?
- 30) Explain how the organization of ground tissue in stems differs between monocots and dicots, and analyze how these differences reflect their evolutionary adaptations to different environmental conditions.
- 31) Examine the relationship between the external morphology and internal anatomy of the epidermis in plants. How do specialized structures like trichomes and cuticle contribute to plant survival in diverse environments?

SECTION D: Case-based/Competency (4 marks each) (2 × 4 = 8)

- 32) 1 How does the organization level of Ctenophores contribute to their ability to exhibit bioluminescence compared to Porifera? (2 marks)
- 33) 2 Using the data provided, analyze why deep-water ctenophores show higher bioluminescence intensity across all stimuli compared to surface-dwelling specimens, and explain the evolutionary significance of this difference. (2 marks)

Case-Based Question 2 (4 marks)

Comparative Leaf Anatomy Study

A botanist conducted a study comparing the anatomy of leaves from plants growing in different environmental conditions. The microscopic examination revealed the following features:

Sample A: Leaf with thick cuticle, sunken stomata primarily on lower surface, multiple layers of palisade parenchyma, compact arrangement of cells with minimal intercellular spaces, and presence of bulliform cells.

Sample B: Leaf with thin cuticle, stomata evenly distributed on both surfaces, single layer of palisade parenchyma, loosely arranged spongy parenchyma with large intercellular spaces, and absence of bulliform cells.

The botanist also measured the rate of transpiration and photosynthesis in both samples under identical laboratory conditions:

Parameter	Sample A	Sample B
Transpiration rate (mmol H ₂ O/m ² /s)	1)2	3)5
Photosynthetic rate (μmol CO ₂ /m ² /s)	12)5	8)7

Based on this case study, answer the following questions:

SECTION E: Long Answer (5 marks each) (3 × 5 = 15)

Long Answer Questions with Internal Choice (5 marks each)

Question 1

Describe the phylum Ctenophora in detail, highlighting its distinctive features and evolutionary significance. Explain how ctenophores differ from other diploblastic animals in terms of their organization and physiological processes.

OR

Compare and contrast the cellular level of organization seen in Porifera with the tissue level of organization in Coelenterata. Discuss how these different levels of organization reflect evolutionary advancement in the animal kingdom, with specific examples from each phylum.

Question 2

Analyze the morphological and anatomical differences between monocotyledonous and dicotyledonous plants with respect to their roots, stems, and leaves. Explain how these structural differences are related to their physiological functions and environmental adaptations.

OR

Describe the structure of a dorsiventral leaf in detail with the help of a labeled diagram. Explain how its anatomical features are adapted for the process of photosynthesis and gas exchange, and how these features differ from those of an isobilateral leaf.

Question 3

Explain the morphology and anatomy of the digestive and excretory systems of a frog. Discuss how these systems are adapted to support the amphibious lifestyle of frogs, allowing them to survive both in aquatic and terrestrial environments.

OR

Describe the circulatory and respiratory systems of frogs in detail. Explain how these systems are integrated to facilitate efficient gas exchange during different life stages and environmental conditions. Discuss the significance of the three-chambered heart in amphibians from an evolutionary perspective.

QUESTION PAPER GENERATION SUMMARY

Class: 11

Subject: Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

Section Details:

Section A (MCQs): 14 questions | 1 mark each | Total: 14 marks

Chapters: 4, 5 | Context: 27643 chars

Section B (Very Short Answer): 10 questions | 2 mark each | Total: 20 marks

Chapters: 5, 6 | Context: 26654 chars

Section C (Short Answer Type I): 7 questions | 3 mark each | Total: 21 marks

Chapters: 6, 7 | Context: 27717 chars

Section D (Case-based/Competency): 2 questions | 4 mark each | Total: 8 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars

Section E (Long Answer): 3 questions | 5 mark each | Total: 15 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars